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Computationally Efficient Multivariate Changepoint Detection with Subsets

Rebecca Killick
r.killick@lancs.ac.uk
Joint work with Ben Pickering & Sean Ryan
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Outline

- Motivation
- Current Approaches

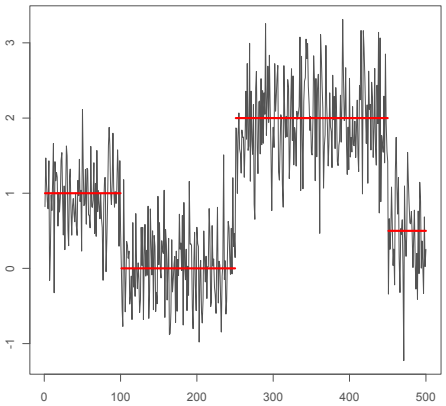
- Modelling Subset-Multivariate Changepoints
- Subset Multivariate Optimal Partitioning
- Example

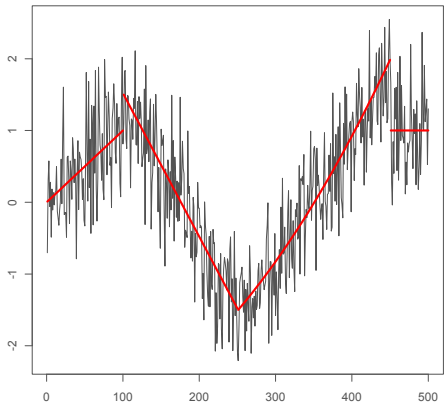
- Approximate but computationally efficient implementation
- Simulations
- Examples

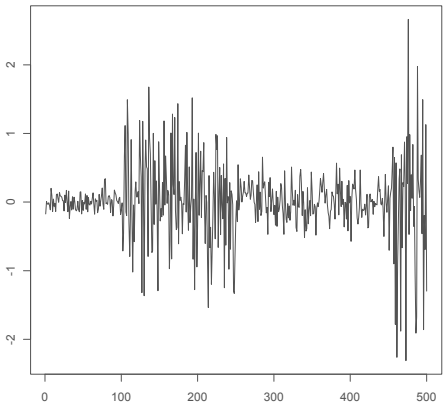
Scene setting: What is changepoint analysis?

What are changepoints?

For data $y_1, \ldots, y_n$, a changepoint is a location τ where the statistical properties of $y_1, \ldots, y_\tau$ are different from $y_{\tau+1}, \ldots, y_n$ in some way.





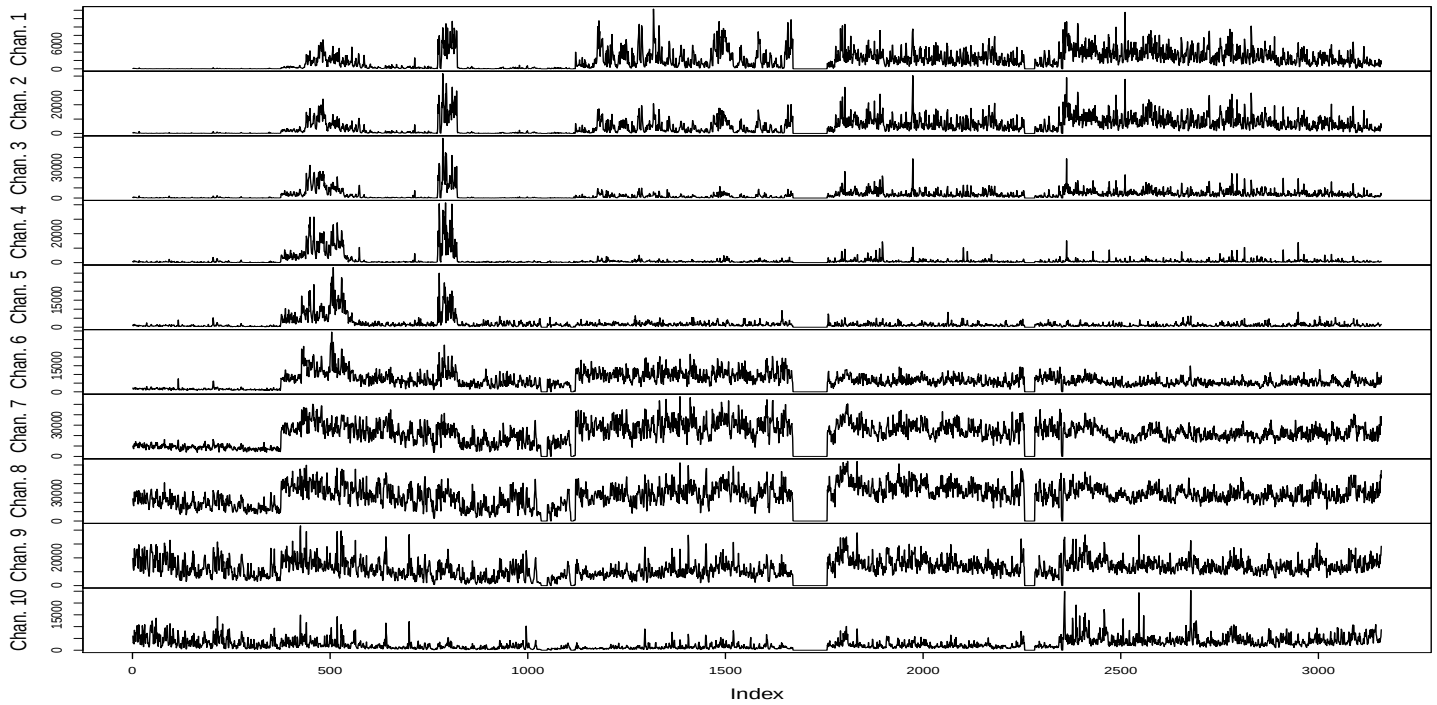


This talk is general methodology $\Longrightarrow$ no restriction on the change.

Motivation: Distributed Acoustic Sensing

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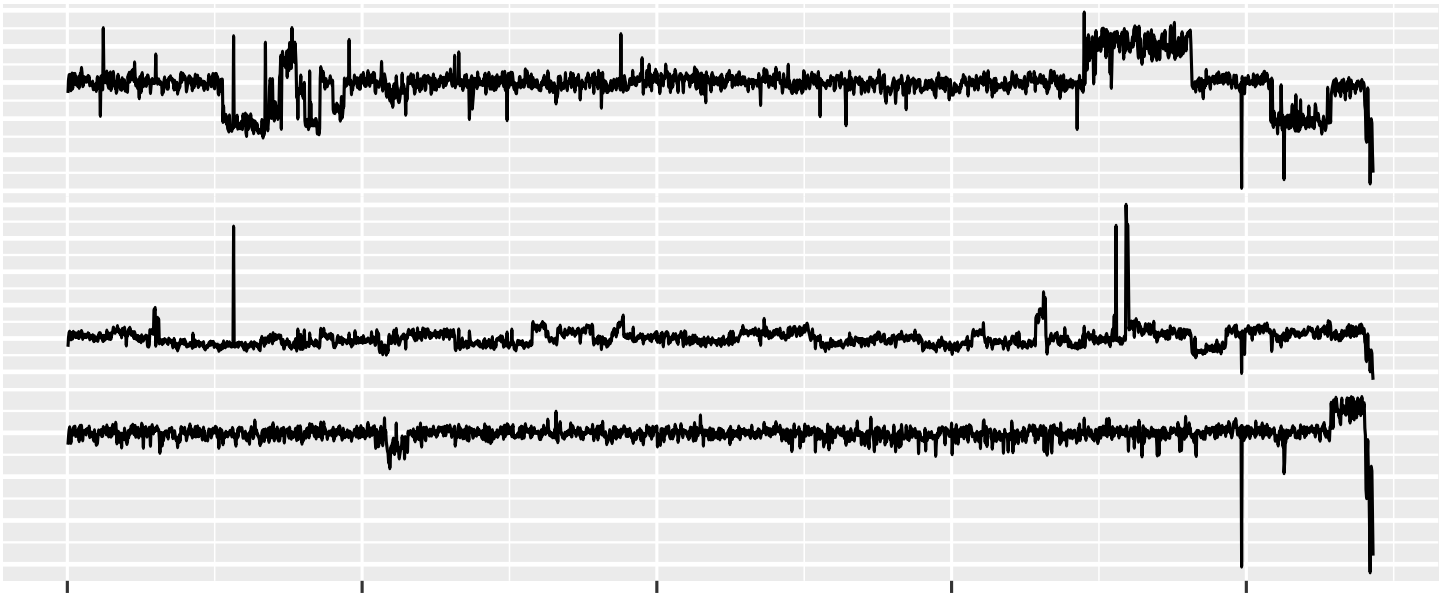
- Vibration measurements from oil well - multiple depths over time
- Fourier-transformed data (used by engineers), sub-sampled 10x3159



- Changes in vibration can indicate presence of impurity slugs
- Errors in measuring equipment cause striping
- True changes likely to affect only subsets

Motivation: Genomics

- CGH measurements on DNA 43x2215
- Want to detect abnormalities common across patients with a disease



Aim

Explicitly (and **efficiently**) identify *both* **changepoint locations** and corresponding **affected variable subsets**.

In moving to a multivariate setting a number of different scenarios can arise:

- The process in each channel could be unconnected to the rest (i.e. repeated use of univariate cpt methods might be appropriate);
- There may be some shared structure across channels. For example
  - Changes occur at the same time in all channels;
  - Changes occur in a subset of channels at the same time.
- The nature of the change could vary from one channel to another;
- . . . and doubtless many more scenarios!

Current Approaches

Naive Approach: Repeated Univariate Application

Apply univariate method independently to each variable

- ✔ Explicitly identifies affected variables (but not necessarily correct)
- ✘ Omits multivariate power $\implies$ miss changes with small magnitude

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Fully-Multivariate Methods

Apply fully-multivariate method to entire (multivariate) series

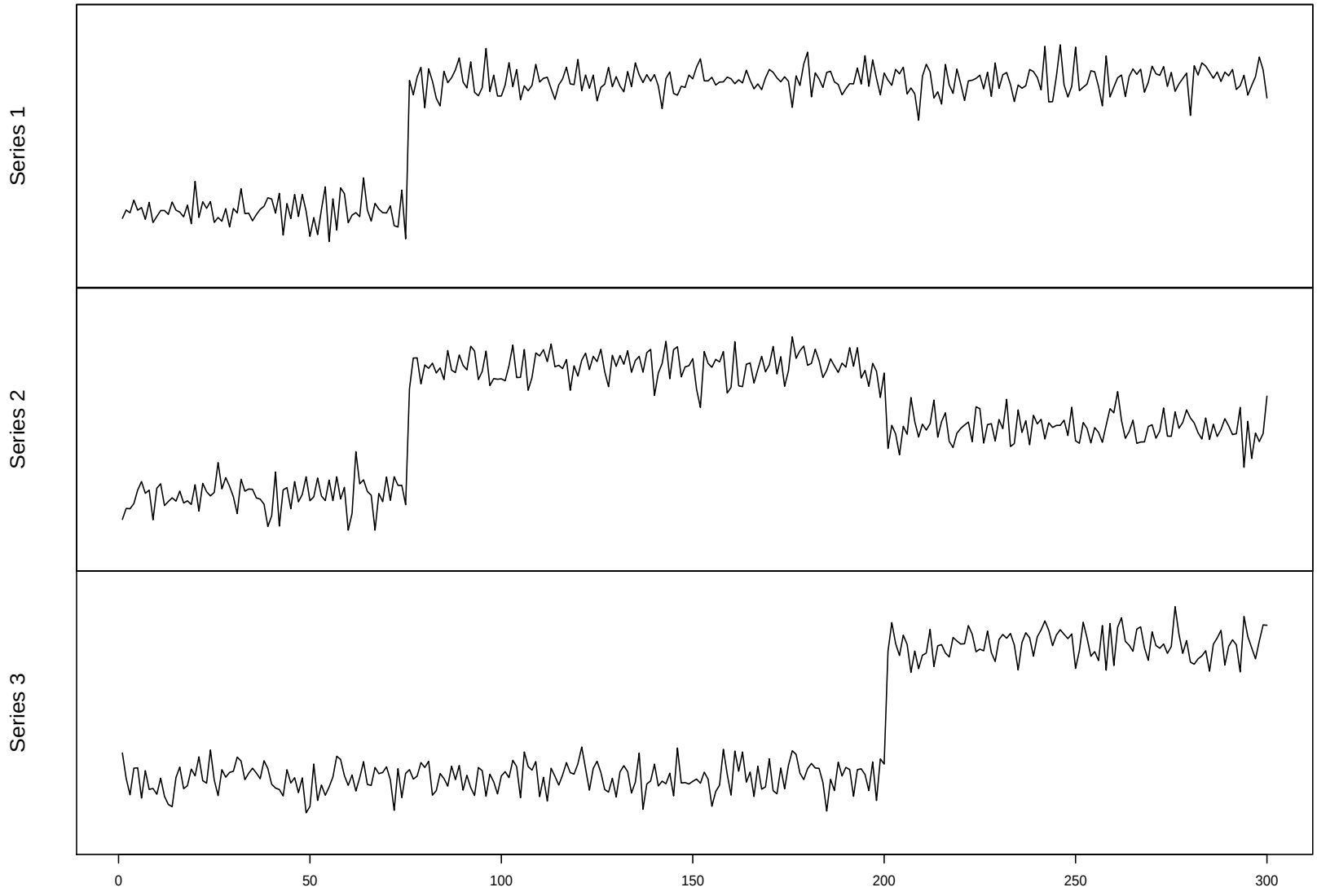
- ✔ Pool information from multiple channels $\implies$ increased detection power
- ✘ Does not explicitly identify subsets (assumes all variables affected)
- ✘ Variables with no change just add noise

Our Framework

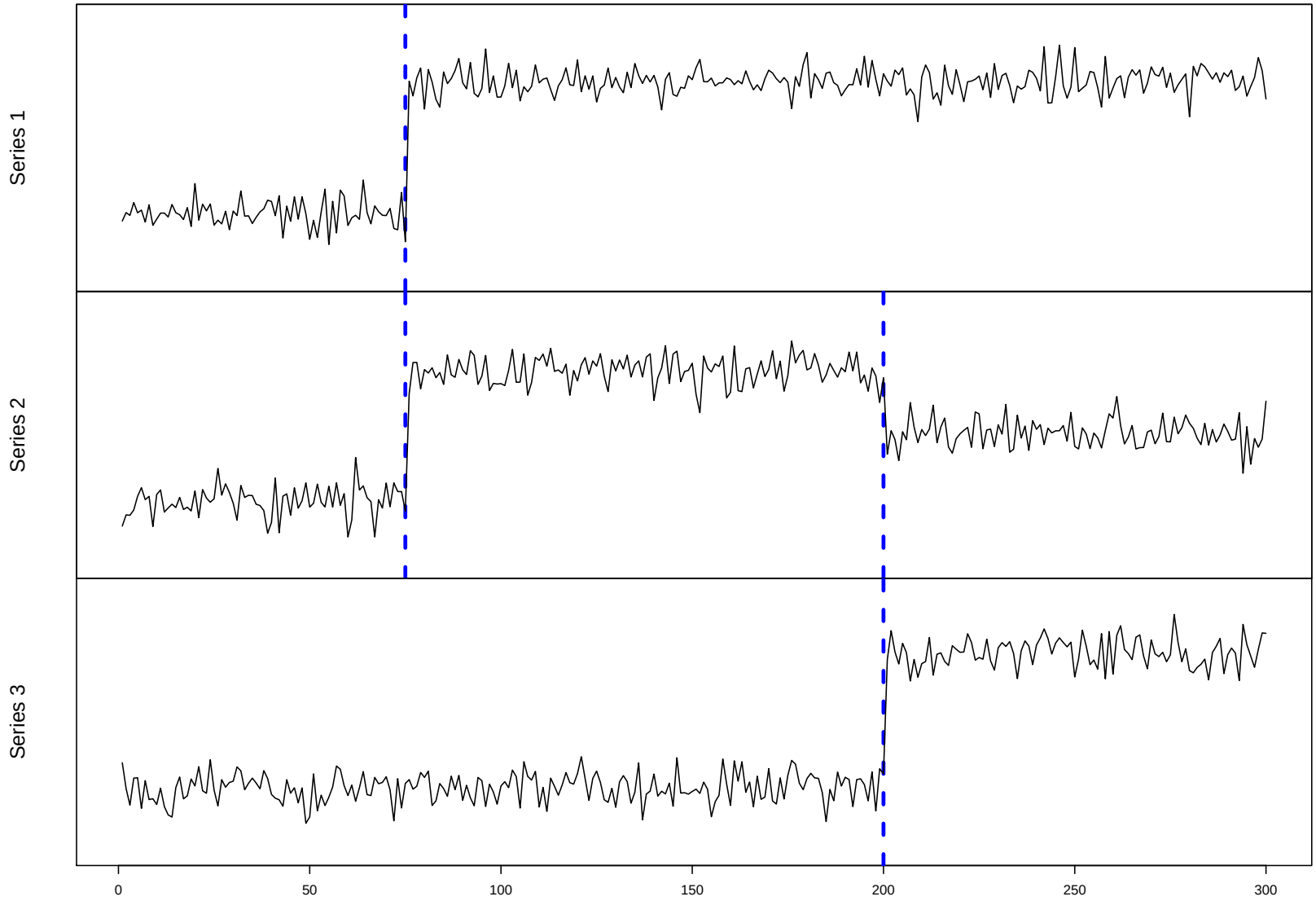
Consider a situation where the data observed within a channel are iid and there is no explicit dependence (coherence) between channels.

- Model changepoints and affected variables using single entity: **changepoint vectors**
  - Allows for segmentation of series involving subset-multivariate changes
- Formulate penalised cost for subset-multivariate segmentation using **dual penalty concept**
  - Allows separate control over univariate and multivariate detection power
- Minimise penalised cost using **Optimal Partitioning**
  - Automatically optimises number, locations and subsets of changepoints.

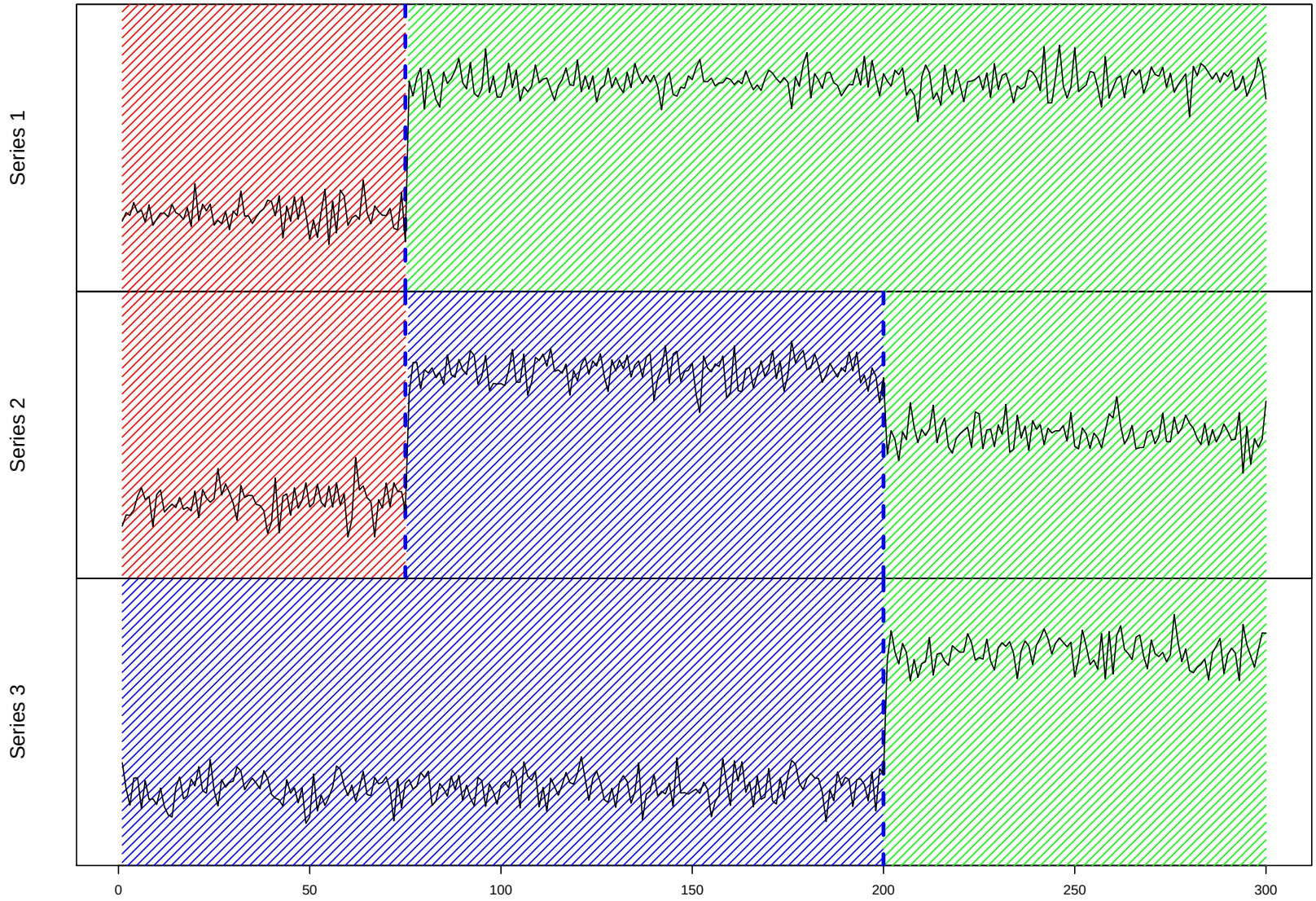
Modelling Subset-Multivariate Cpts



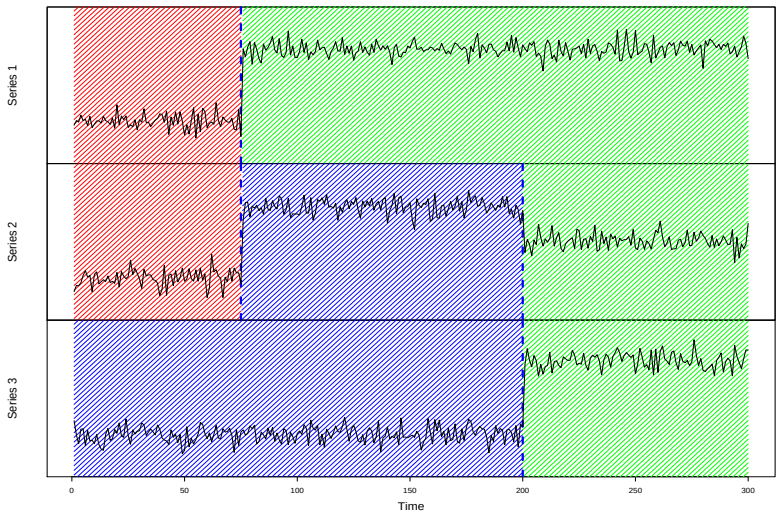
Modelling Subset-Multivariate Cpts



Modelling Subset-Multivariate Cpts



Modelling Subset-Multivariate Cpts



Changepoints:

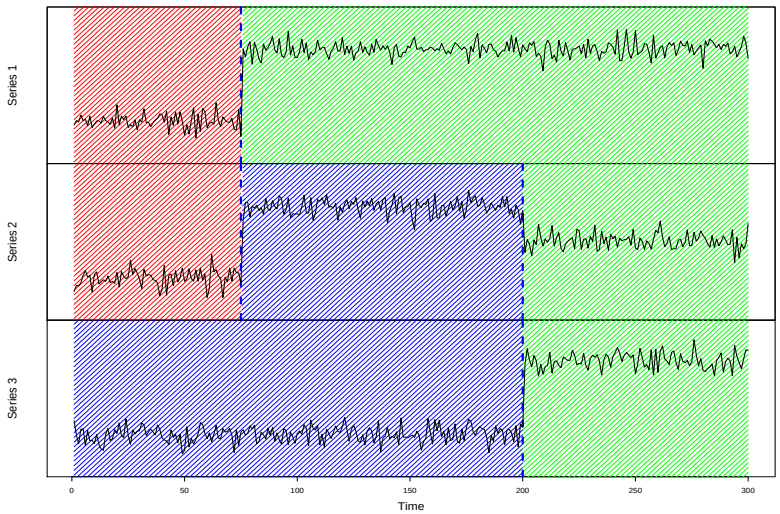
$\tau_1 = 75 \quad \tau_2 = 200 \quad \tau_3 = n = 300$

Changepoint Vectors:

$c_{\tau_1} = (75, 75, 0) \qquad c_{\tau_2} = (75, 200, 200)$

$c_{\tau_3} = (300, 300, 300)$

Modelling Subset-Multivariate Cpts



Changepoints:

$\tau_1 = 75 \quad \tau_2 = 200 \quad \tau_3 = n = 300$

Changepoint Vectors:

$c_{\tau_1} = (75, 75, 0) \qquad c_{\tau_2} = (75, 200, 200)$

$c_{\tau_3} = (300, 300, 300)$

Cost of Segmentation (assuming known changepoints)

$$\begin{aligned}\mathcal{C}(\boldsymbol{X}) &= \left[\mathcal{C}(X^1_{(\tau_0+1):\tau_1}) + \mathcal{C}(X^2_{(\tau_0+1):\tau_1})\right] + \left[\mathcal{C}(X^2_{(\tau_1+1):\tau_2}) + \mathcal{C}(X^3_{(\tau_0+1):\tau_2})\right] \\ &\quad + \left[\mathcal{C}(X^1_{(\tau_1+1):\tau_3}) + \mathcal{C}(X^2_{(\tau_2+1):\tau_3}) + \mathcal{C}(X^3_{(\tau_2+1):\tau_3})\right] \\ &= \sum_{k=1}^{m+1}\sum_{j=1}^{p} \mathbb{I}(c^j_{\tau_k} = \tau_k)\mathcal{C}\left(X^j_{(c^j_{\tau_{k-1}}+1):c^j_{\tau_k}}\right).\end{aligned}$$

Benefit: Encapsulates info on changepoints *and* affected subsets

Subset Multivariate Optimal Partitioning (SMOP)

Aim to minimise:

$$\sum_{k=1}^{m+1}\left\{\sum_{j=1}^{p}\left[\mathbb{I}(c^j_{\tau_k}=\tau_k)\left(\mathcal{C}_j\left(X^j_{(c^j_{\tau_{k-1}}+1):c^j_{\tau_k}}\right)+\alpha\right)\right]+\beta\right\}$$

$\mathcal{C}_j$ – Channel-Specific Cost Function

- Assigns cost to univariate data segment

α – Penalises increase in number of channels included in subset.

β – Penalises increase in total number of changepoints.

SMOP: Dynamic Program

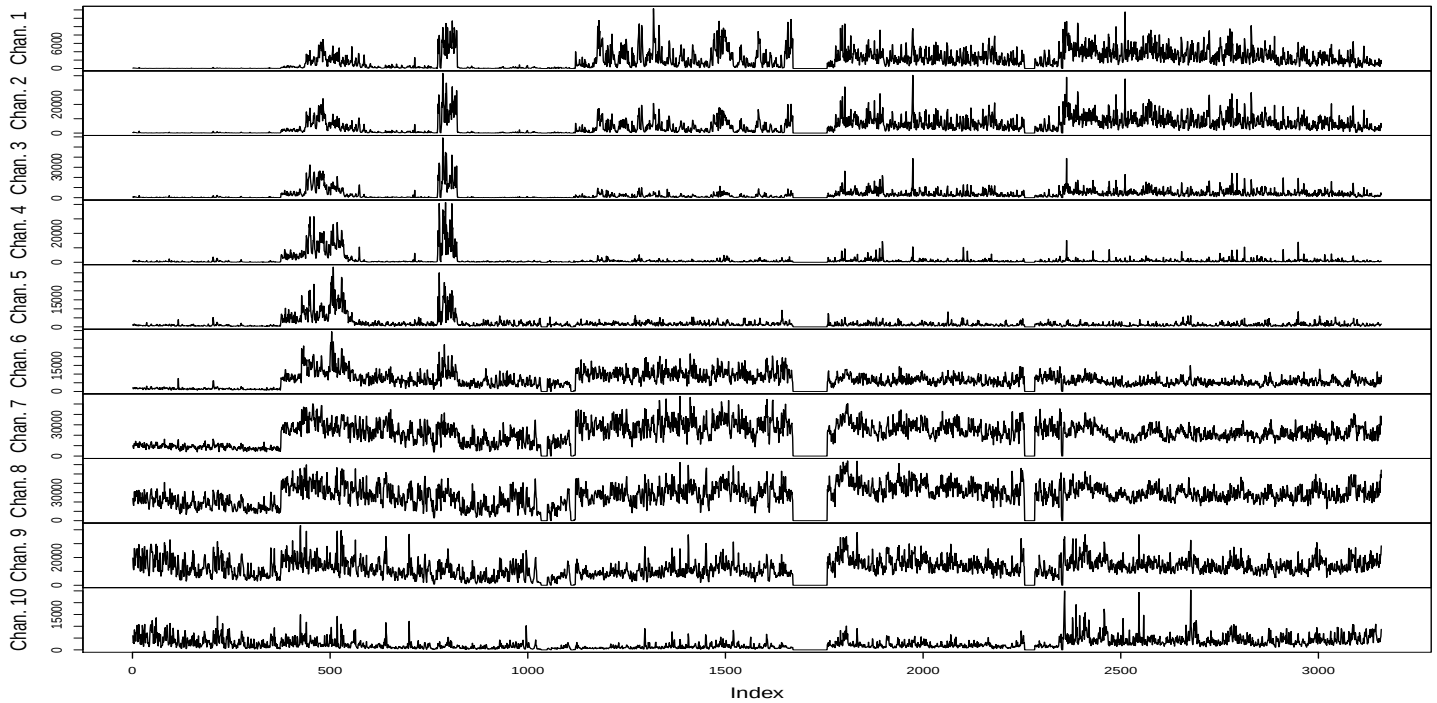
This formulation allows us to adapt optimal partitioning (dynamic programming) for this setting $\implies$ some speed up possible.

For a given changepoint vector $c_u \in C_n$ (where $u = \max(c_u)$), we have

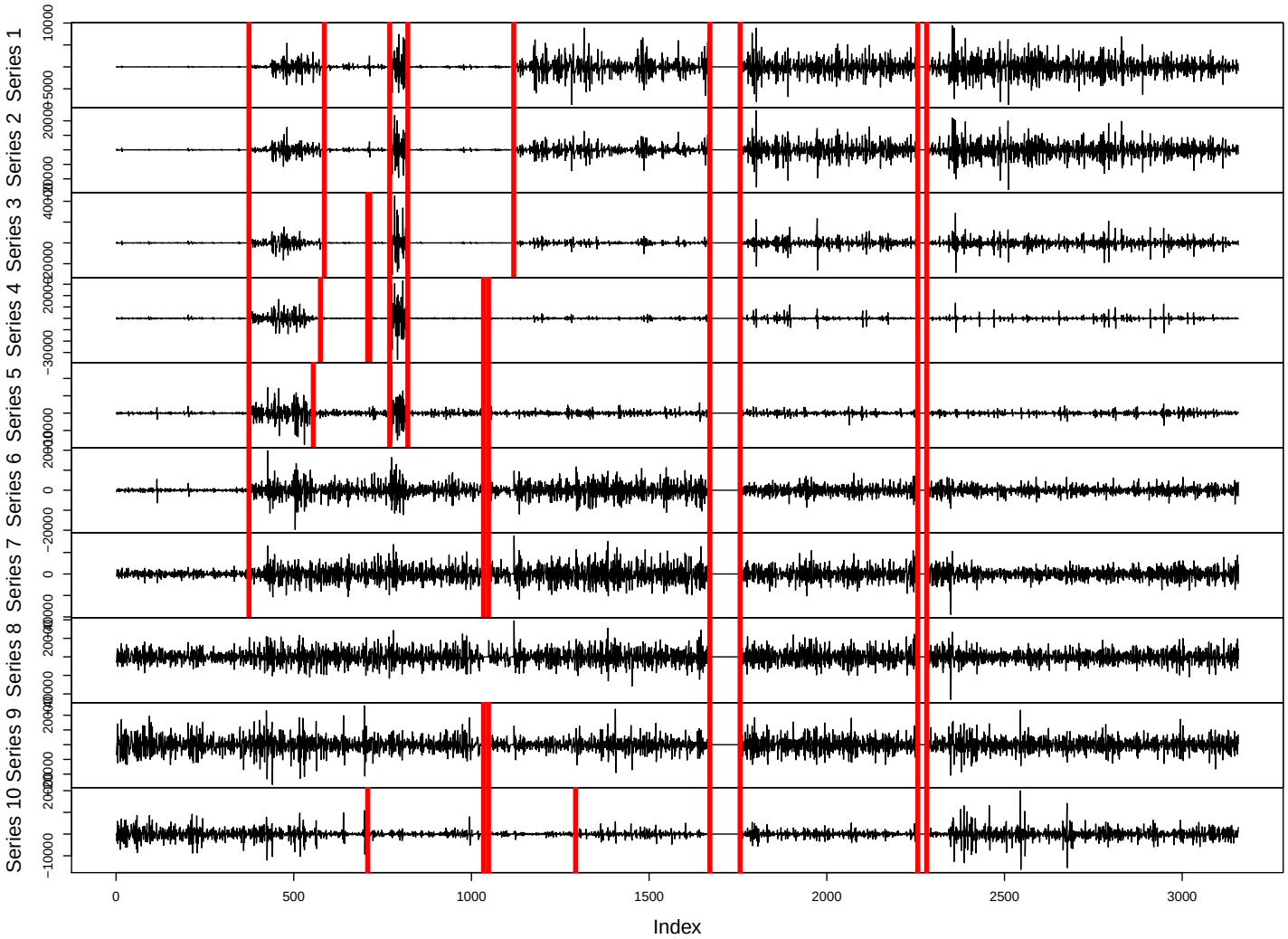
$$F(c_u) = \min_{0 \leq t < u} \left\{ \min_{c_t \in \{\bar{C}_t \,:\, c_t^j \leq c_u^j \;\forall\, j\}} \left[F(c_t) + \sum_{j=1}^{p} \left[\mathbb{I}(c_t^j \neq c_u^j) \left(\mathcal{C}_j(X^j_{(c_t^j+1):c_u^j}) + \alpha \right) \right] + m(c_t, c_u)\beta \right] \right\}.$$

Acoustic Sensing Data

- Vibration measurements from oil well – multiple depths over time
- Fourier-transformed data (used by engineers), sub-sampled

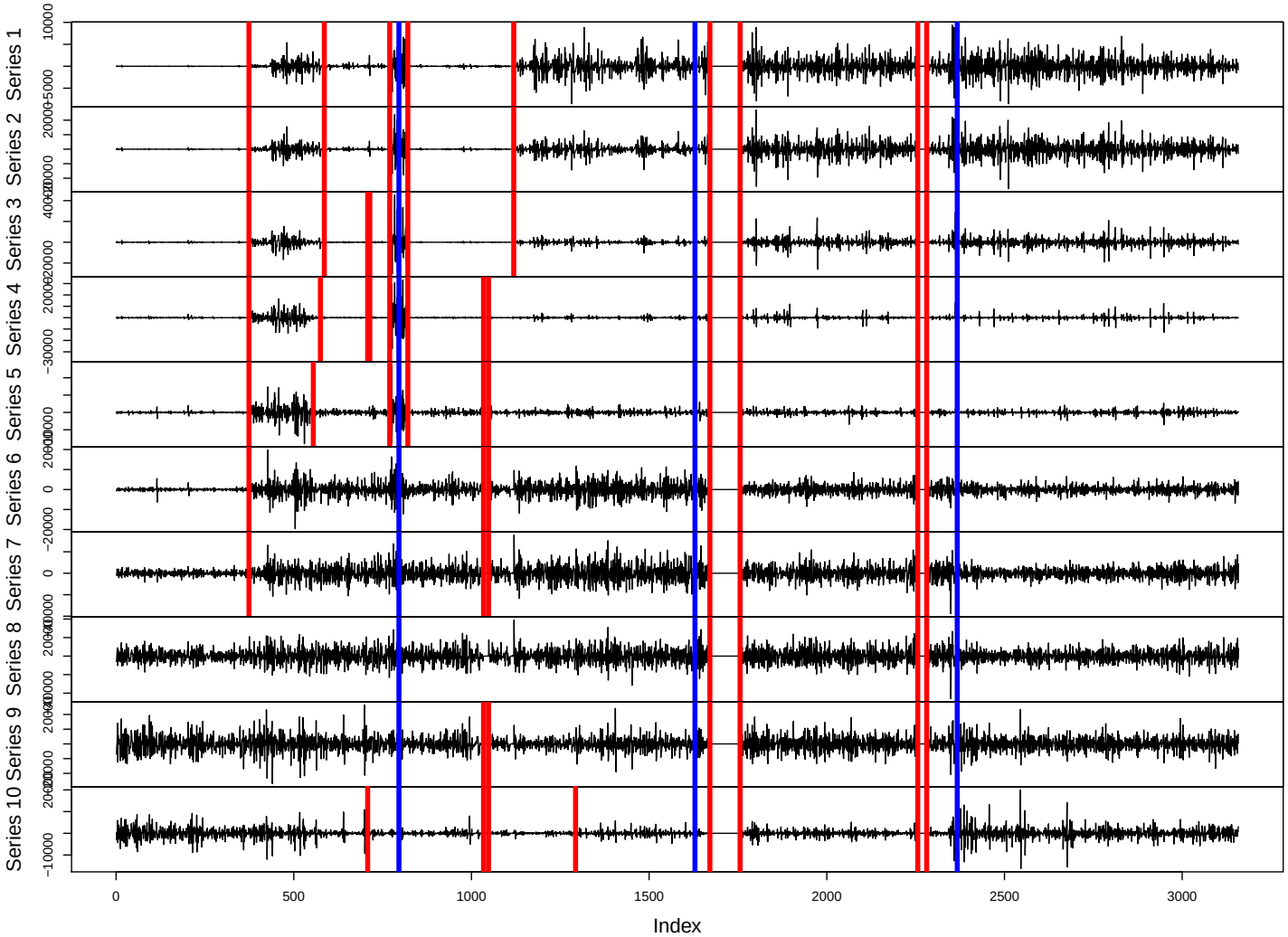


Acoustic Sensing Data



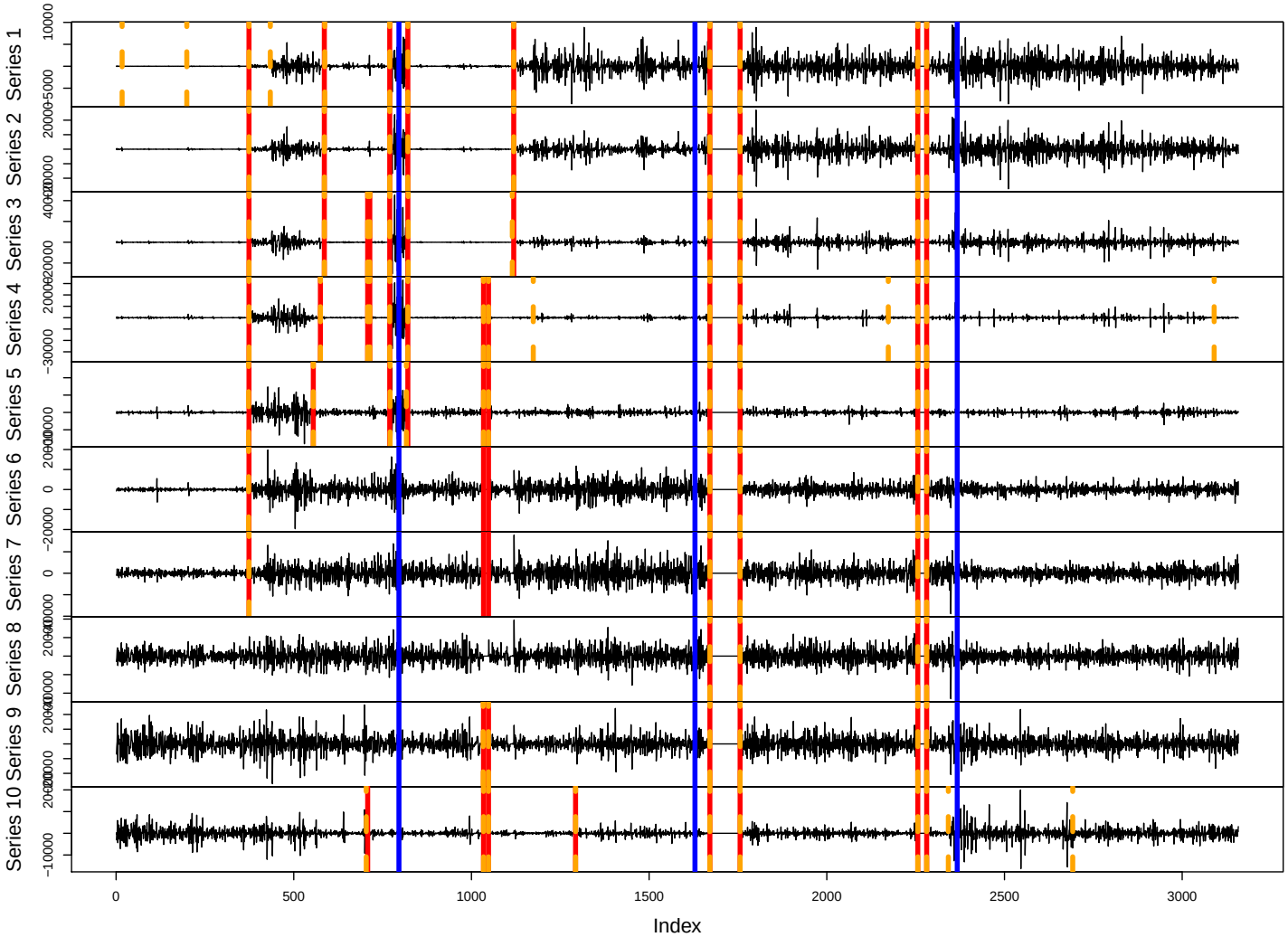
RED: SPOT

Acoustic Sensing Data



RED: SPOT BLUE: e-cp30

Acoustic Sensing Data



RED: SPOT BLUE: e-cp30 ORANGE: univariate PELT.

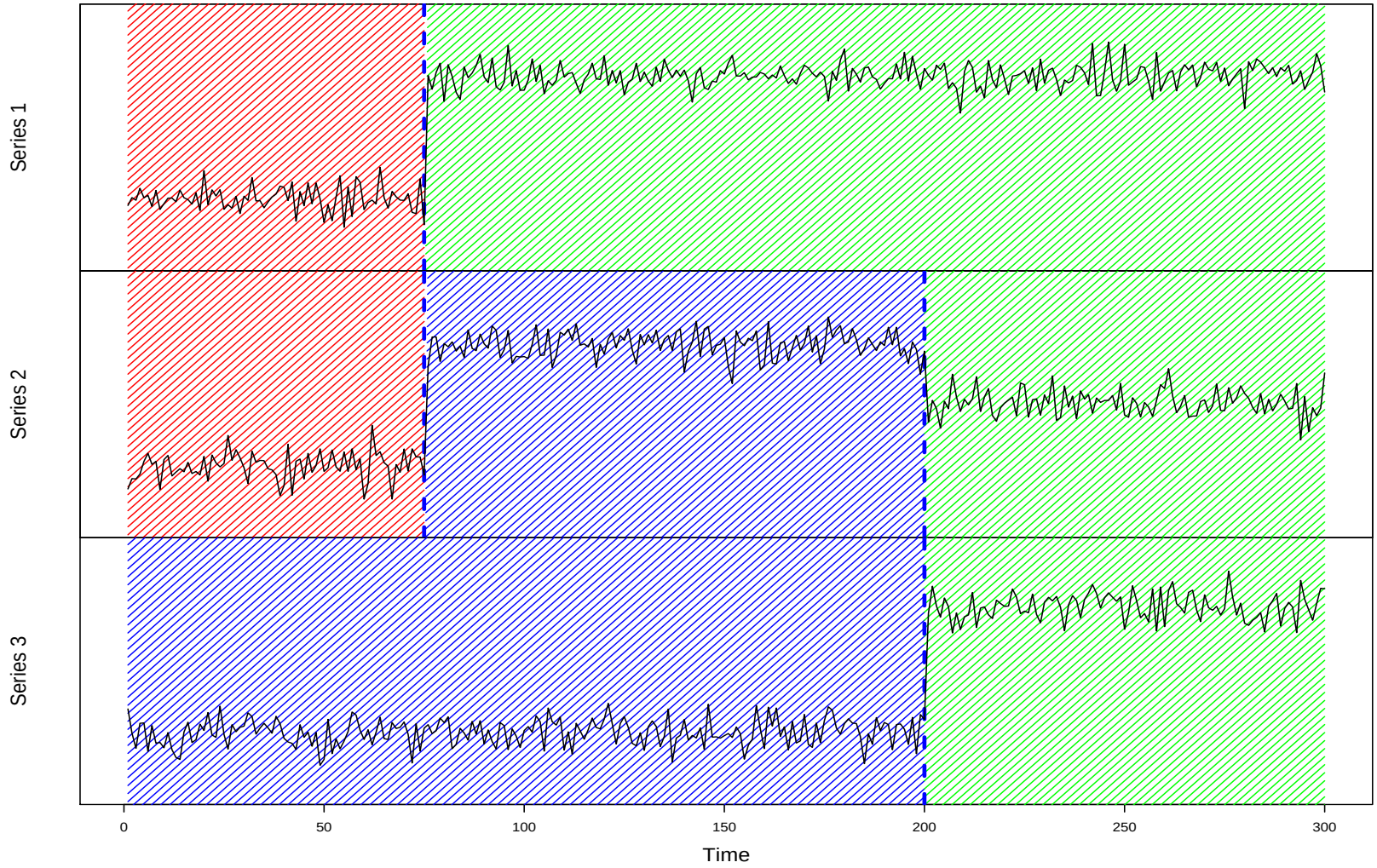
SMOP: computational considerations

- For an exact changepoint approach, we must evaluate this penalised cost function for **all** possible changepoint locations and subsets
  $\implies$ computationally intense algorithm!
- Hence with Optimal Partitioning the **exact** approach is $\mathcal{O}(pn^{2p})$.

SPOT

Approximate but computationally efficient

Approximation of SMOP



Windowed Cost - Univariate

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$$\hat{\mathcal{C}}(p, q, w_i) = \begin{cases} \mathcal{C}(X_{p:q}|\theta(p, p + w_i)) & \text{for} \quad p - q \geq w_i \\ \mathcal{C}(X_{p:q}|\theta(p, q)) & \text{for} \quad p - q < w_i. \end{cases}$$

Proposition

Assume that $m_0 \leq m_U$, $\theta_r^0 \neq \theta_{r+1}^0$ $(1 \leq r \leq m_0)$ and $(\tau_r - \tau_{r-1})/n$ converge to q_r for $1 \leq r \leq m_0$ as $n \to \infty$. Then $Pr(\hat{m} = m_0) \to 1$ as $n \to \infty$.

Windowed Cost - Multivariate

Natural window choice: $w_i = \tau_i - \tau_{i-1}$

$$\hat{F}(n) = \sum_{k=1}^{m+1} \left\{ \sum_{j=1}^{p} \left[\mathbb{I}(S_k^j = 1) \left(\mathcal{C}_j \left(X^j_{(\tau_{k-1}+1):\tau_k} | \hat{\theta}^j_k \right) + \alpha \right) \right. \right.$$
$$\left. \left. + \mathbb{I}(S_k^j = 0) \left(\mathcal{C}_j \left(X^j_{(\tau_{k-1}+1):\tau_k} | \hat{\theta}^j_{k-1} \right) \right) \right] + \beta \right\}$$

$$\hat{F}(s) \approx \min_{1 \leq t < s} \left\{ \hat{F}(t) + \hat{C}(t, s | \theta_t) + \beta \right\}$$

Overall cost of this is $\mathcal{O}(pn^2)$.

SPOT-Pruning

Works in the same way as PELT pruning (Killick et al., JASA 2012). BUT in the multivariate setting the pruning is not as efficient as in univariate.

Pruning Rule

$$\hat{F}(t) + \hat{C}(t, s) - p\alpha \geq \hat{F}(s)$$

Then s will always be a better fit than t for any time $T > s$.

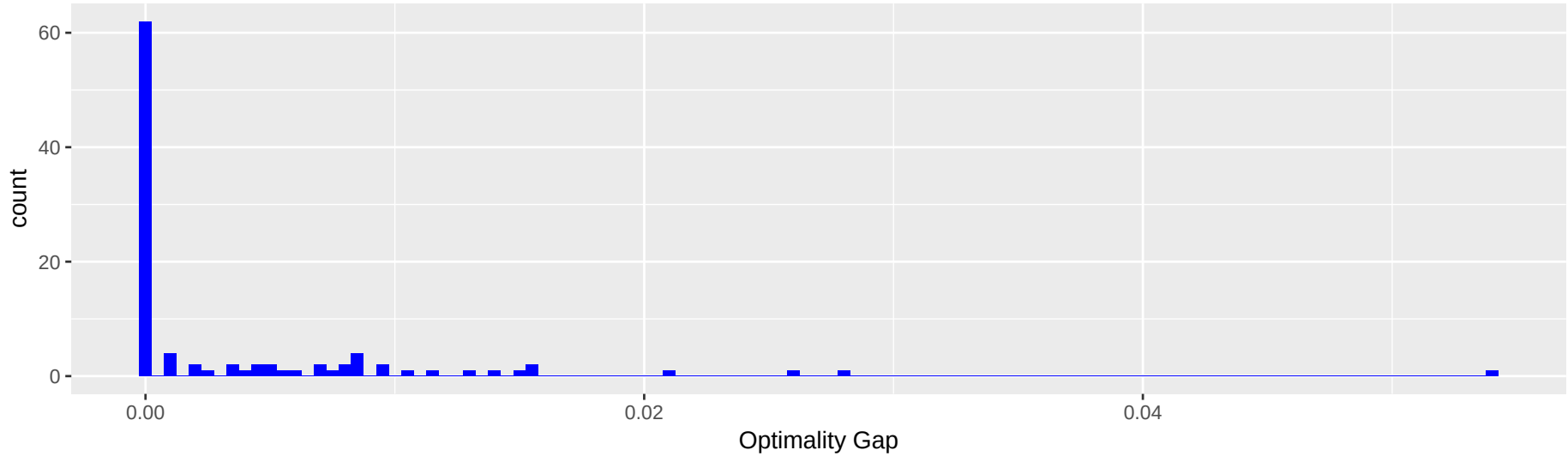
Theoretically we can show that for fixed p the algorithm is $\mathcal{O}(n)$.

SMOP/SPOT Comparison

Setup:

- Change in Normal mean.
- $n = 100$, $p = 3$
- 2 changes ($m = 2$)
- magnitude of change is random inside $[0.8, 1.3]$.

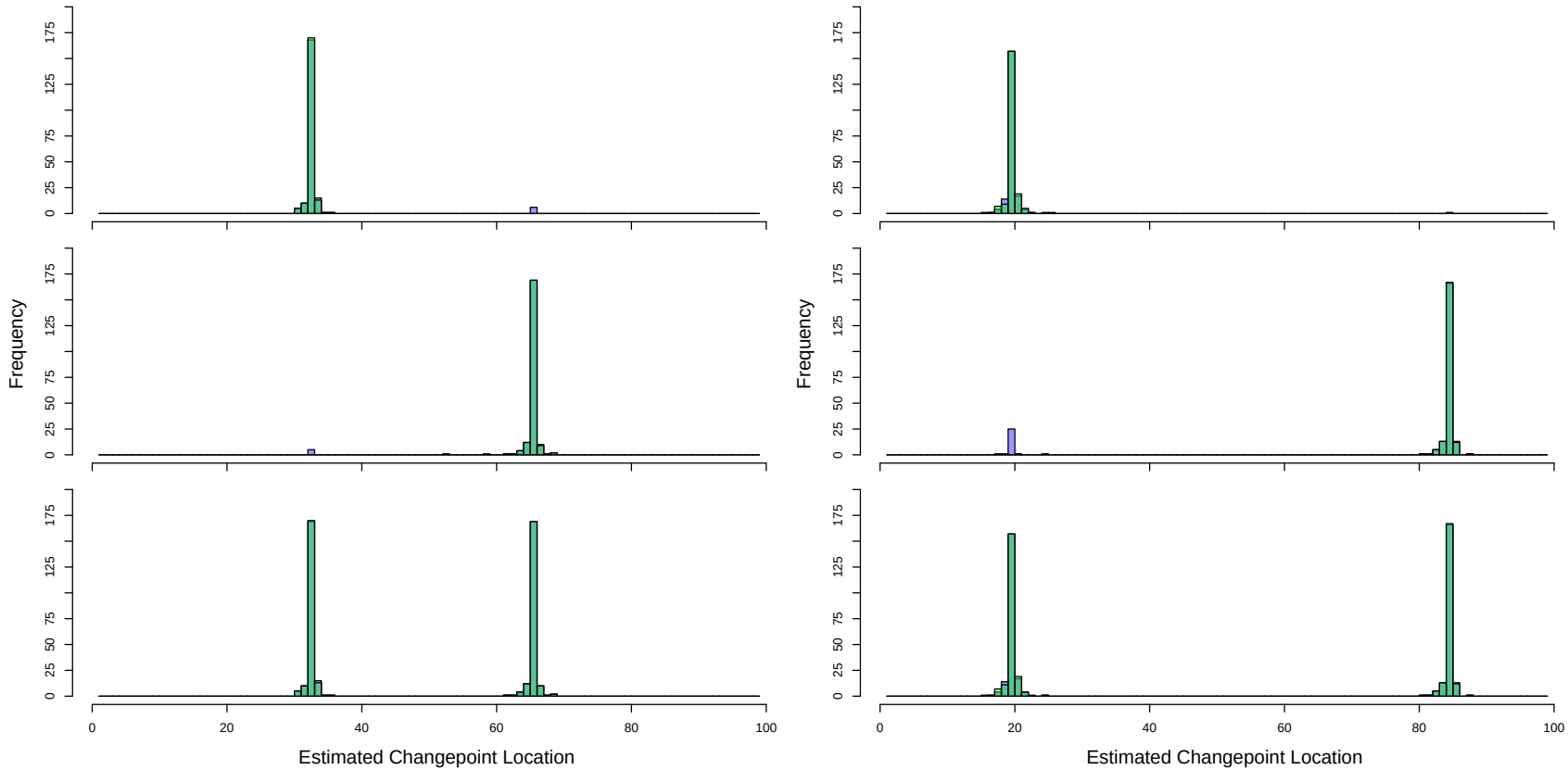
Method	SPOT	SMOP
MSE	**0.17**(0.15)	0.21(0.16)
Time (s)	**0.07**(0.01)	15030.10(1329.5)



SMOP/SPOT Comparison

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Left: Changes at 33 and 66 - almost indistinguishable
Right: Changes at 20 and 85 - demonstrating the approximation



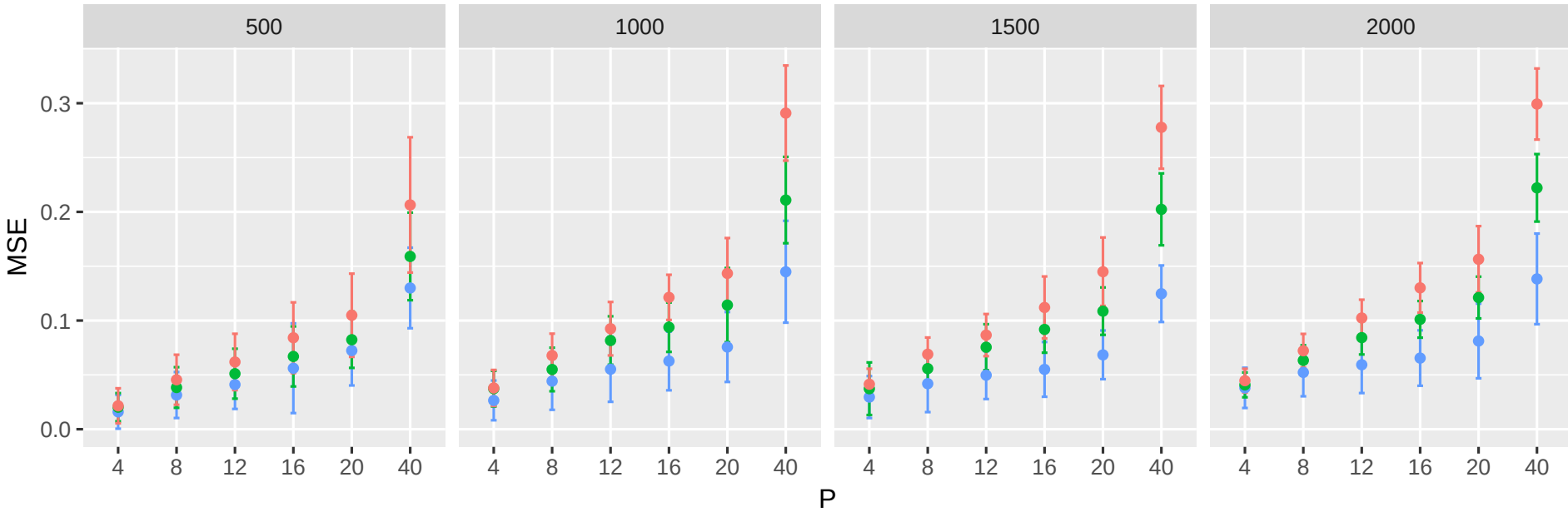
Simulation Examples

- Change in mean (so we can compare with Inspect)
- m increases with n, $E(\tau_{i+1} - \tau_i + 1) = 200$
- Expected number of affected variates is 0.2, 0.5, or 0.8 with equal probability

Compare with the following fully multivariate approaches:

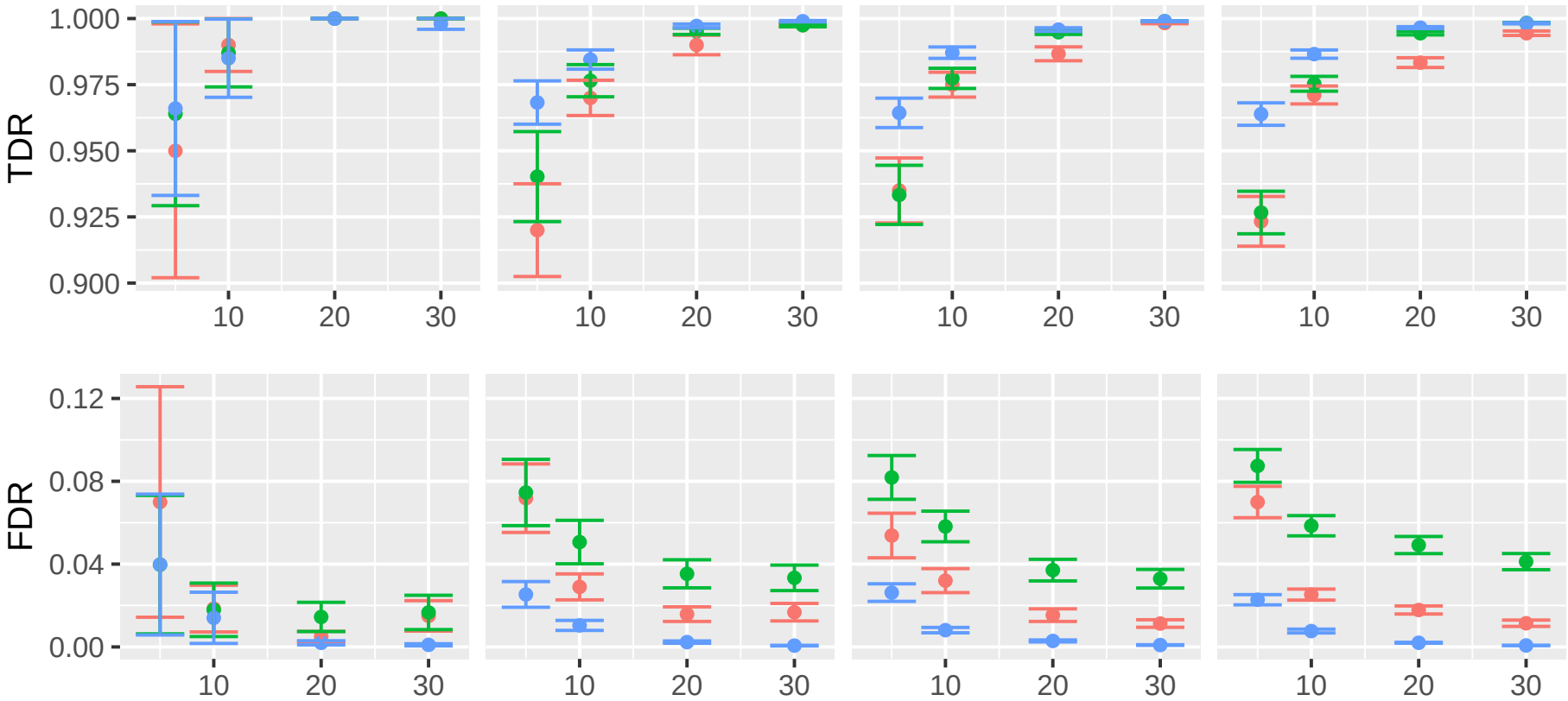
- ECP, non-parametric change in energy
- Inspect, restricted to changes in mean.

Simulation Examples



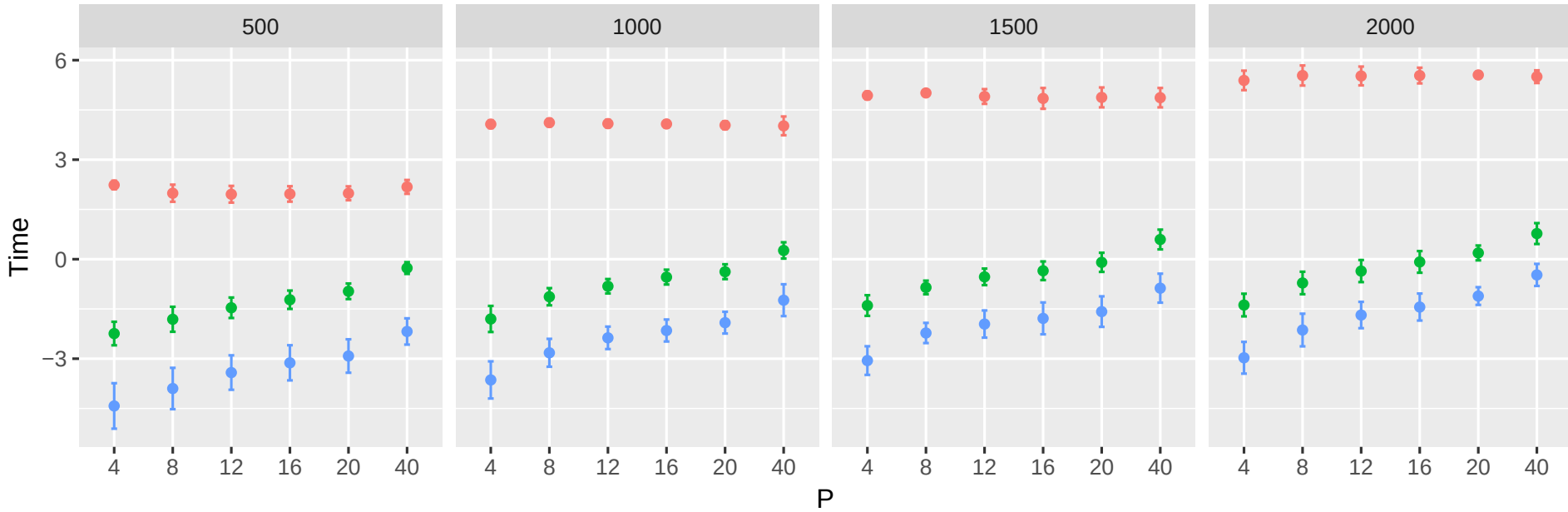
Red=ECP, Green=Inspect, Blue=SPOT

Simulation Examples



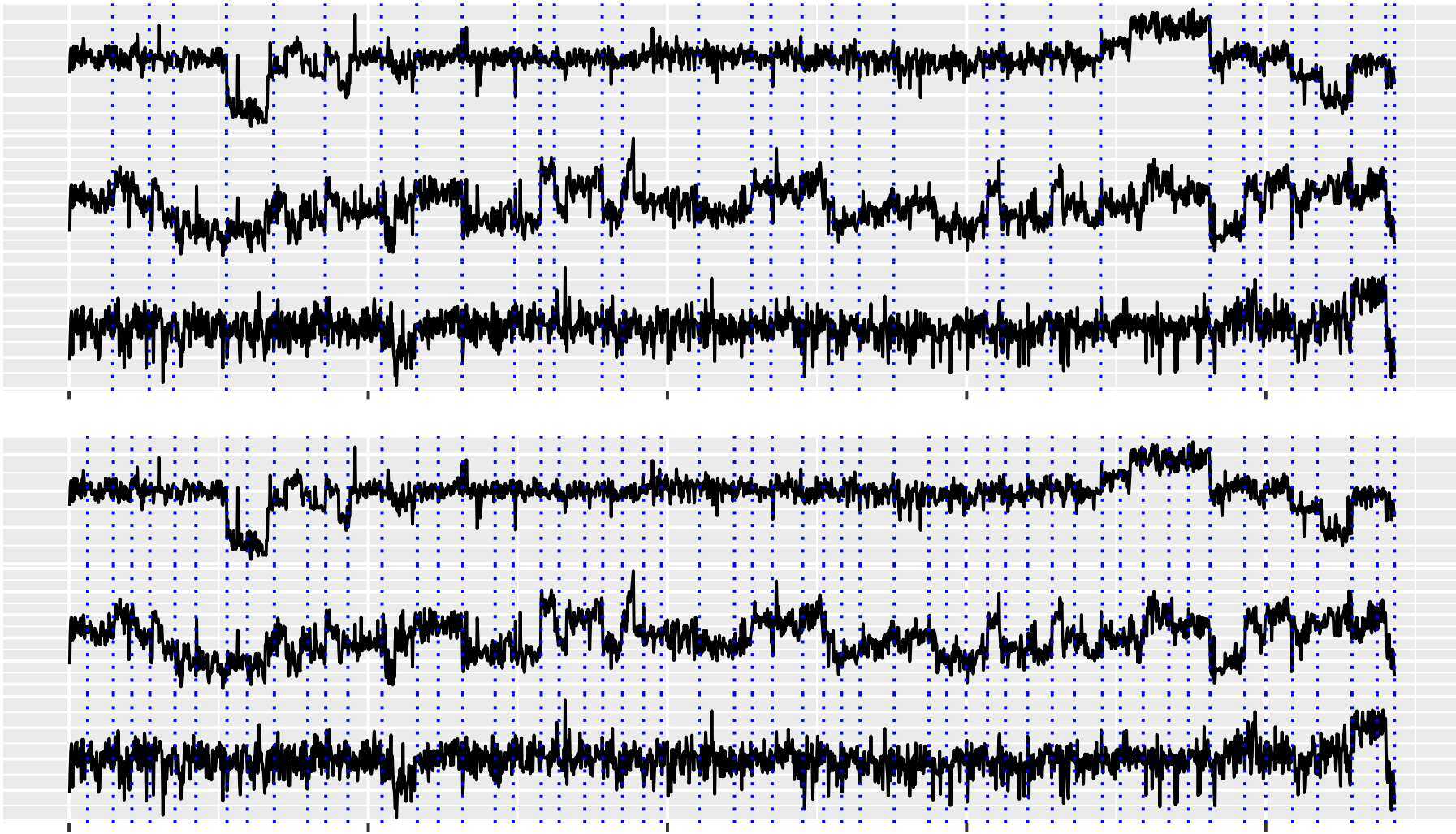
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Simulation Examples

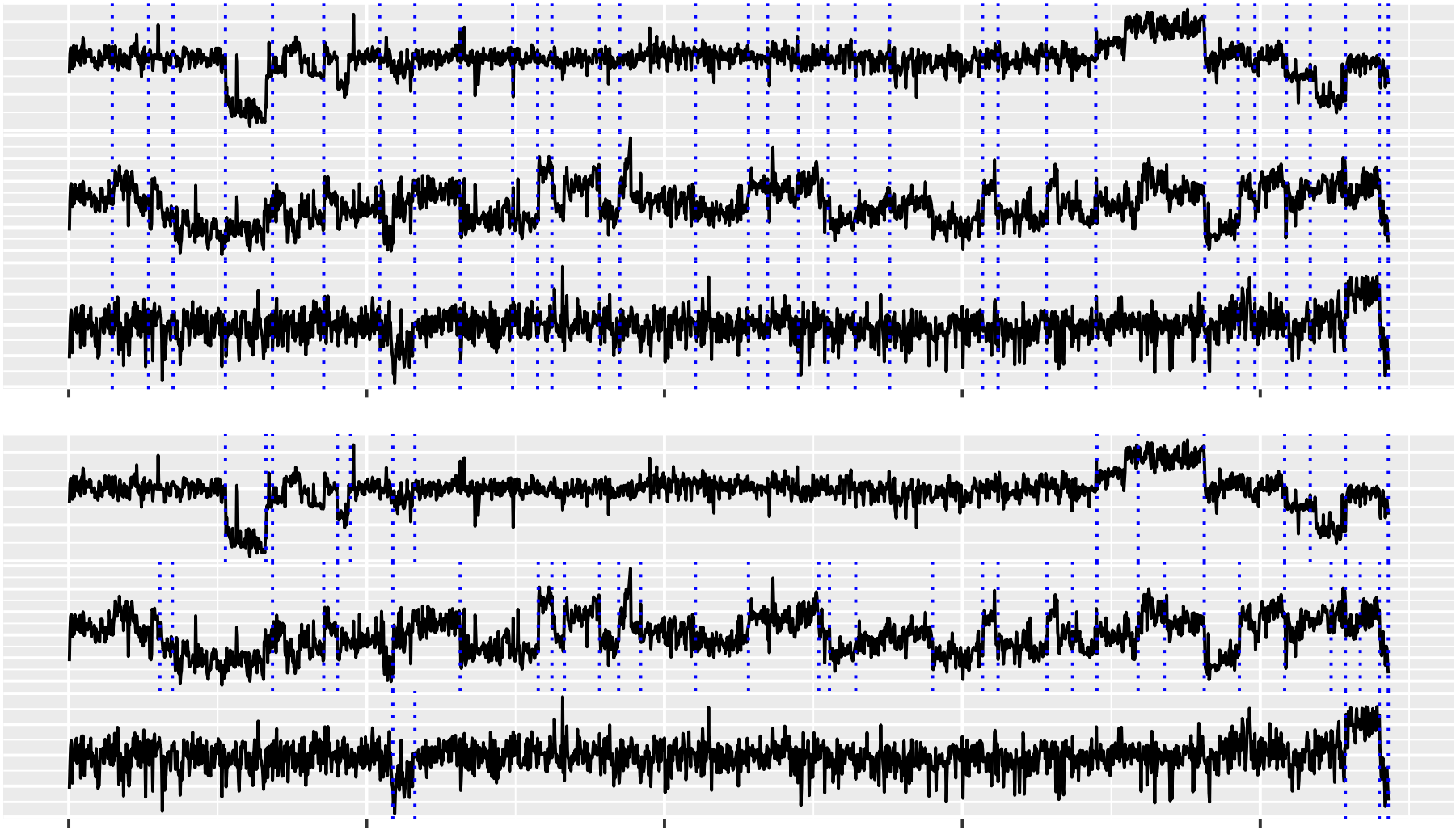


Red=ECP, Green=Inspect, Blue=SPOT

CGH results: Inspect & ECP



CGH results: Inspect & SPOT



CGH results

- Inspect paper: 30 most significant changepoints
- ECP default: 54 changepoints
- SPOT default: 69 changepoints

- But for SPOT only 10 affect more than 25 series out of 43

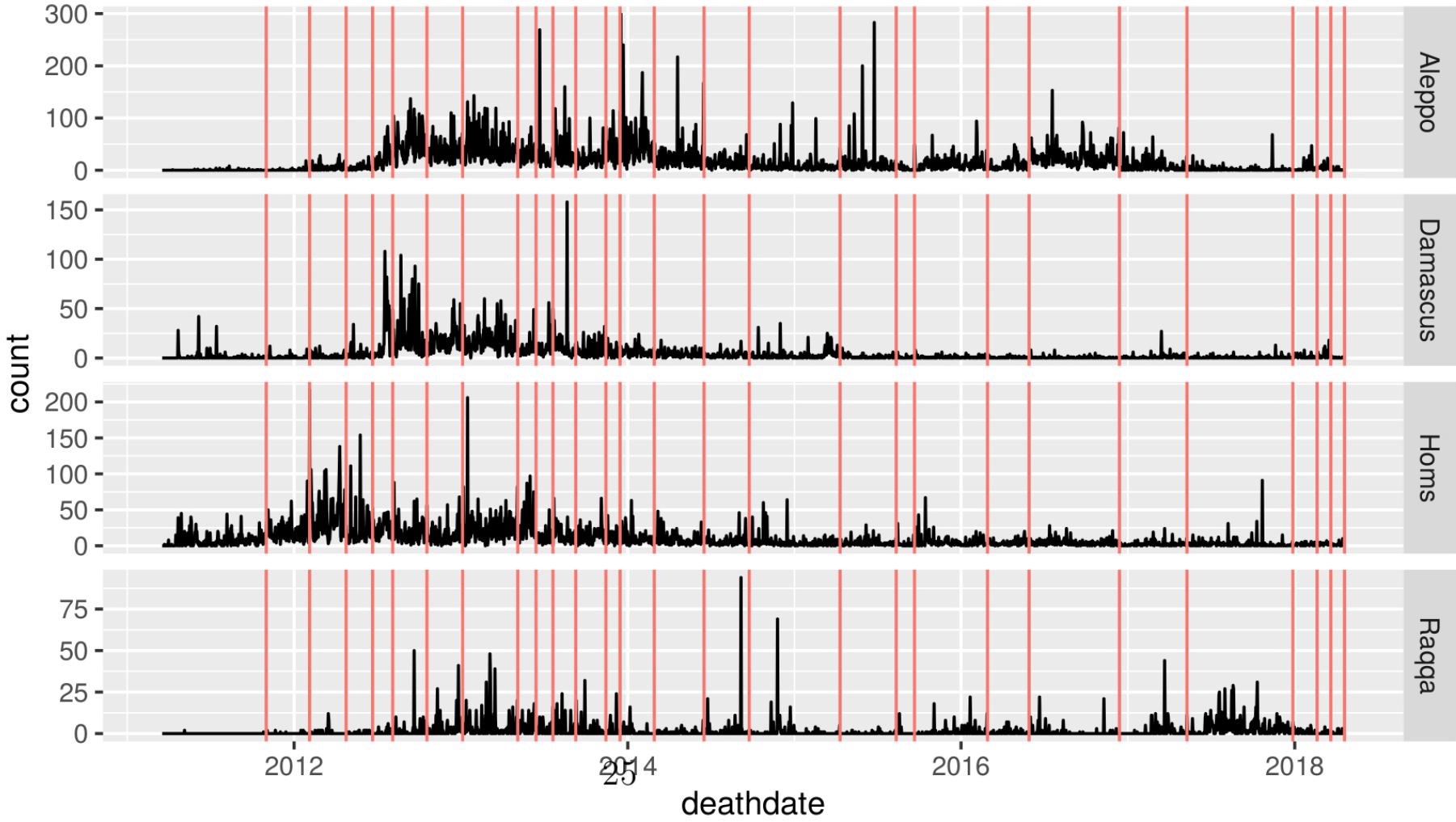
Syrian Civil War

- Violations Documentation Centre
- Records confirmed deaths

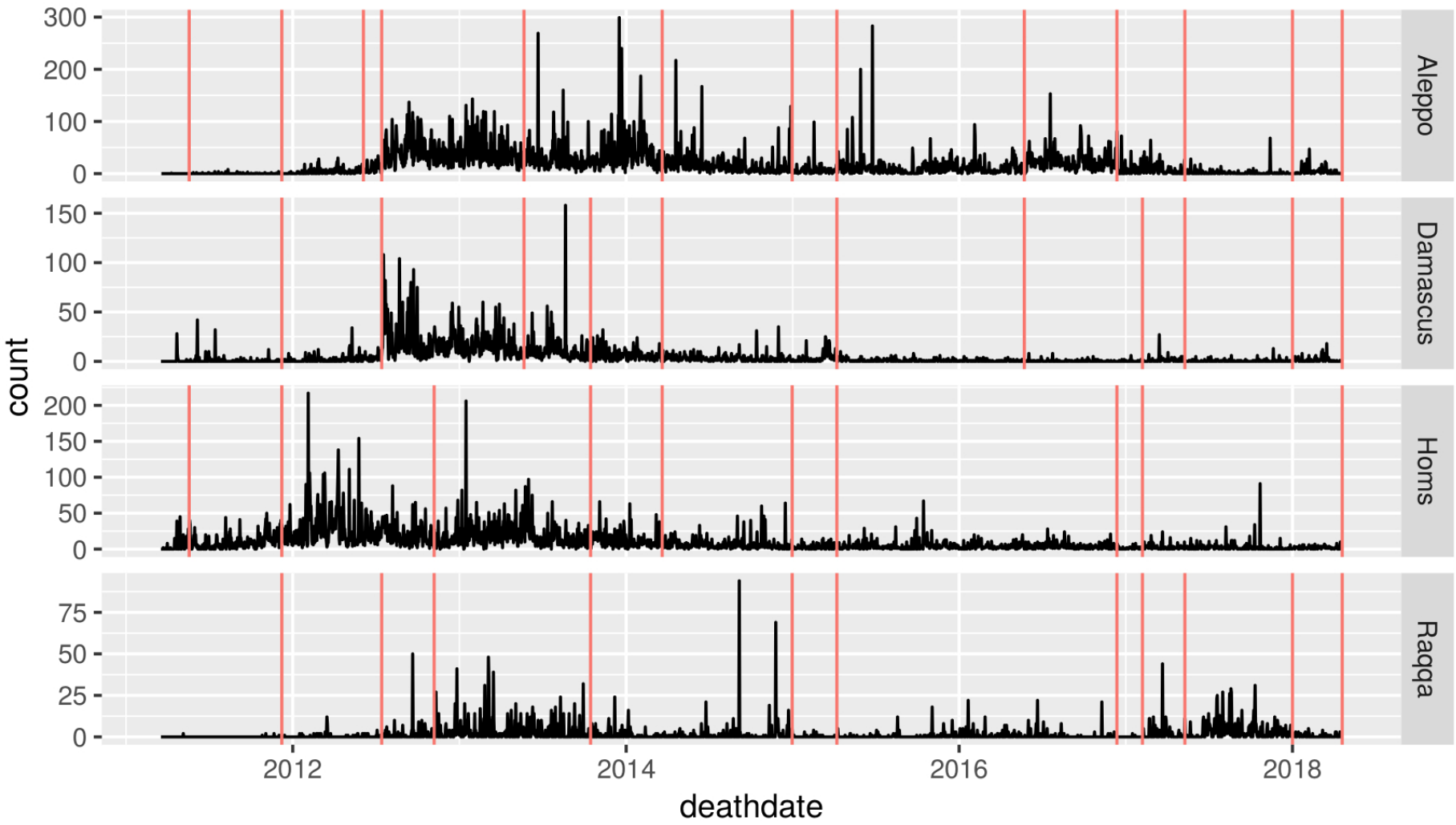
- Name of victim
- Date and region where they died

- Discrete, bounded support

Syrian results: ECP



Syrian results: SPOT



Summary

- Identifying which subsets of multivariate series undergo a change is important in many practical applications
- SMOP: A promising subset multivariate changepoint detection framework
- SPOT: using windows to approximation SMOP permits good segmentation accuracy in comparatively much lower computational time
- SPOT and SMOP available in `changepoint.mv` R package to be released on CRAN shortly

- Invite others to use the SMOP framework and devise either approximations or computationally efficient methods for exact minimizations.